The Reggae Shark Protocol (v1.3.1 — Reviewer-Audited Edition): 41 Lean 4 Theorems, Pinned-Key Ed25519 Ledger Verification, 200-Bit Interval Patankar Bridge & Bandlimited LRSC

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Repository: github.com/reggaesharkk/The-Reggae-Shark-Protocol · Concept DOI: 10.5281/zenodo.23618032

DOIs: 10.5281/zenodo.23635351 (v4) · 10.5281/zenodo.23624605 (v3) · 10.5281/zenodo.23620074 (v2) · 10.5281/zenodo.23618033 (v1)

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October 25, 2026

Abstract

The Reggae Shark Protocol (v1.3.1) is a verifiable execution kernel and multi-cloud AI gateway that binds agent prompt dispatches to four verification layers: (1) an append-only, SHA-256 hash-chained and RFC 8032 Ed25519-signed **IntentSeal Verifiable Research Kernel (VRK v0.3.1)** ledger requiring an explicit private seed (`SHARK_ED25519_SEED_HEX`), mandatory pinned `expected_pubkey` verification, append-only `fsync` persistence, signed head records (`intentseal-vrk-head-v1`) to detect tail truncation, an N-way associative Merkle-DAG join-semilattice, and optional child-process key isolation (`IsolatedSignerSubprocess`); (1B) a thread-synchronized **Request-Lifecycle Typestate and Worker-Epoch Isolation** state machine (`TypestateLifecycle.lean`, 15 zero-`sorry` theorems) verified by differential trace testing against its Lean `#eval` JSON export; (2) a bounded-LRU per-tenant **4D Soft-Wall ODE Governor** $(\gamma, H_{\text{obs}}, \dot{I}, \Psi)$ backed by 17 4D boundary theorems (`ExistenceVerification.lean`), 9 3D soft-wall theorems (`BoundaryVerification.lean`), and **200-bit outward-rounded** `mpmath.iv` **Patankar–Euler interval enclosures**, plus an illustrative 120-segment Duhamel parameter study on `data/wp19_segments.json`; and (3) a **Bandlimited LRSC** operator ($M = 99, K = 11$) with linear perturbation budget scaling (`perturbation_budget_scaling`).

1 Four-Layer Verification Architecture & Provenance (41 Lean 4 Theorems)

Every request submitted to `MultiProviderGateway.process_request` or the local loopback HTTP proxy (`POST http://127.0.0.1:8000/v1/execute`) traverses four deterministic verification gates:

Layer	Module / Artifact	Enforced Mathematical & Cryptographic Invariant
Layer 1	`shark_protocol/kernel.py` `SECURITY.md` · `AUDIT_LOG.md`	**Mandatory Pinned-Key Ed25519 VRK v0.3.1 Ledger & Signed Head:** Anchored at trust root `f010a99d...05a0d0f` (10.5281/zenodo.23018459); requires `cryptography` and explicit `SHARK_ED25519_SEED_HEX` (no default seed), requires `expected_pubkey` in `verify_chain_list`, writes append-only `fsync` records, and signs `head_hash` commitments.
Layer 1B	`lean/TypestateLifecycle.lean` `data/lean_transition_table.json`	**Typestate, Epoch & Differential Trace Suite (15 Theorems) + RLock:** 6-state request automaton (`UNINIT` $\to \cdots \to$ `ZEROED_FREED`) with absorbing `ZEROED_FREED`, inductive trace safety (`runTrace`), and differential equivalence between Lean `#eval` JSON and Python `RequestSlot`.
Layer 2	`lean/ExistenceVerification.lean` `lean/BoundaryVerification.lean`	**4D Soft-Wall ODE Governor ($\mathcal{B}_4$, 26 Theorems) & 200-Bit Interval Bridge:** Adaptive RK4, `float64` Patankar–Euler, and 200-bit `mpmath.iv` interval enclosure (`step_ode_patankar_mp_interval`) inside $\mathcal{B}_4 = [0, 10] \times [0, 100] \times [0, \dot{I}_{\max,4D}] \times [0, 1]$.
Layer 3	`LRSCCompressor` (`kernel.py`)	**Bandlimited LRSC Odd-Ring Projection ($11/99 = 1/9$):** Projects context onto the $M = 99$ odd-ring filter V_{ring} ($\sigma_{\min}(S_{11}) = 0.181646 > 0$, 10.17605/OSF.IO/NM5BW), bounds perturbation scaling under assumed $L \geq 0$ ([6]), and binds distinct compressed/uncompressed SHA-256 digests.
Layer 4	`shark_protocol/gateway.py`	**Fail-Closed Multi-Tenant Gateway & `ThreadingHTTPServer`:** Enforces waking drive $\mathcal{P}_0 \geq 1.0 \gg 10/99$, token-bound tenants (`derive_tenant_from_token`), bounded LRU (`max_tenants=128`), rate limits (HTTP 429), header-only API keys (`x-goog-api-key`), and HTTP 413 caps.

Table 1: Four-layer verification architecture of The Reggae Shark Protocol (v1.3.1).

Canonical Lean 4 Module Provenance & CI Axiom Gate (41 Zero-`sorry` Theorems): Specifically, `lean/BoundaryVerification.lean` (SHA-256: `49229b61...e417`, 9 theorems) is byte-for-byte identical to Physics v1.5 Appendix A (10.5281/zenodo.23534449), and `lean/ExistenceVerification.lean` (SHA-256: `eec94868...bcae`,

17 theorems) is byte-for-byte identical to *The Theory of Bounded Existence* v1.0.0 (10.5281/zenodo.23608244). Together with the 15 theorems in `lean/TypestateLifecycle.lean`, `lean/AxiomsCheck.lean` and `tools/check_lean_axioms.py` verify in CI that all **41 theorems** depend only on subsets of {`propext`, `Classical.choice`, `Quot.sound`} with zero `sorryAx`.

2 What Is and Is Not Modelled (Epistemic Claim Firewall & Threat Model)

Epistemic Standard (Build. Attack. Falsify. Preserve.): Every theorem and runtime guard in this repository is bounded by explicit domain assumptions, `SECURITY.md` threat boundaries, and `scope.does_not_imply` declarations.

- **Application-Layer Typestate & Subprocess Signer vs. OS Ring-0:** `RequestSlot` and `WorkerEpochGuard` enforce typestate invariants in Python, while `IsolatedSignerSubprocess` isolates the Ed25519 private key in a child OS process (`subprocess.Popen`) across a pipe IPC boundary. They are *not* an OS ring-0 kernel module or hardware HSM. Theorem 14 (`projection_independent_of_secret`) is a definitional projection lemma; process isolation rests on the runtime child-process design and `test_32`.
- **Real Arithmetic in Lean vs. 200-Bit `mpmath.iv` & IEEE-754 Runtime Enclosures:** Lean 4 Theorems 6, 7, 9, 10, 13, and 15 are proved over $\mathbb{R}$; `step_ode_patankar_mp_interval` bridges discrete steps to $\mathbb{R}$ via 200-bit outward-rounded `mpmath.iv` intervals (`INTERVAL_CERTIFIED_RESULT`).
- **Illustrative WP19 Parameter Study vs. Continuum Navier–Stokes:** `evaluate_wp19_piecewise_duhamel_study` evaluates the 120-segment Duhamel recurrence on the assumed segment rates loaded from `data/wp19_segments.json` (10.5281/zenodo.23034545, `NUMERICAL_OBSERVATION`); it does not certify the Galerkin chain while segment rates are assumed, and continuum 3D Navier–Stokes regularity remains `OPEN`.

3 Layer 2 Equations, Core Verification Guards & Audit Remediations (38/38 Tests)

The Layer 2 governor evolves $x(t) = (\gamma(t), H_{\text{obs}}(t), \dot{I}(t), \Psi(t))$ under the smooth (C^∞) soft-wall system [3]:

$$\frac{d\gamma}{dt} = a\dot{I}\mathcal{E}_{\text{avail}}\left(1 - \frac{\gamma}{\gamma_{\max}}\right) - b\gamma, \qquad \frac{dH_{\text{obs}}}{dt} = c\gamma\,(H_{\max} - H_{\text{obs}}) - \dot{I}_{\text{erase}}\tanh\left(\frac{H_{\text{obs}}}{\epsilon_H}\right), \tag{1}$$

$$\frac{d\dot{I}}{dt} = e\tanh\left(\frac{\dot{I}}{\epsilon_I}\right)\frac{dH_{\text{obs}}}{dt} + f\mathcal{P}_{\text{eff}}(\gamma,\Psi) - g\dot{I}, \qquad \frac{d\Psi}{dt} = \kappa_{\text{meta}}R_{\text{self}}(x)\,(1-\Psi) - \mu_\Psi\Psi, \tag{2}$$

where $\mathcal{P}_{\text{eff}}(\gamma,\Psi) = \mathcal{P}_0 + \lambda_{\text{sim}}r_{\text{LRSC}}\gamma + \lambda_{\text{lucid}}\Psi$, $R_{\text{self}}(x) = \text{fb}(x)^2/(\text{fb}(x)^2 + (f\mathcal{P}_0)^2 + \sigma_0^2)$, and $\text{fb}(x) = e\tanh(\dot{I}/\epsilon_I)(dH_{\text{obs}}/dt)$.

1. **Guard 1, Probe A & Track A (200-Bit `mpmath.iv` Patankar–Euler Real-to-Float Bridge):** Updates each coordinate via

$$x_{k+1} = \frac{x_k + \Delta t\,P(x_k)\,x_{\max}}{1 + \Delta t\,(P(x_k) + D(x_k))}, \qquad P(x_k) \geq 0,\ D(x_k) \geq 0,\ 1 + \Delta t(P+D) \geq 1, \tag{3}$$

   paired with Python `float64` floor `FLOAT64_NORMAL_MIN` $= 2.225 \times 10^{-308}$ and `step_ode_patankar_mp_interval` (200-bit outward-rounded `mpmath.iv` intervals of width $< 10^{-45}$ agreeing with `float64` within $< 10^{-12}$, Theorem 13 `patankar_interval_rounding_in_box`, `test_15`, `test_21`, `test_31`).

2. **Guard 2 & Probe B (Pointwise Decay Bound, Token-Bound Tenant LRU, Rate Limit & Waking Drive Floor):** Theorem `psi_decay_floor_bound` proves $d\Psi/dt \geq -\mu_\Psi\Psi$ (yielding $\Psi(t+\Delta t) \geq \Psi(t)e^{-\mu_\Psi\Delta t}$ via Grönwall), while `MultiProviderGateway` binds HTTP tenants to Bearer tokens, enforces `max_tenants=128`, rate-limits requests (HTTP 429), and clamps $\mathcal{P}_0 \geq 1.0 \gg 10/99$ (`test_16`, `test_22`, `test_29`, `test_38`).

3. **Guard 3 & Probe C (Prompt-Committed Bandlimited LRSC Witness & Perturbation Budget Scaling):** `LRSCCompressor.compress` injects a slot-0 anchor to prevent scalar-multiple collisions (`test_27`), projects onto V_{ring} ($\sigma_{\min}(S_{11}) = 0.18164624 > 0$ [4]), evaluates linear perturbation budget scaling $L_{\text{local}}\eta_\perp(\tilde{x})$ on $\|\tilde{x}\|_2 \leq 1$ [6] (Theorem 10 `perturbation_budget_scaling`), and is tested against empirical softmax attention perturbations on random weight matrices (`test_17`, `test_23`, `test_27`, `test_38`).

4. **Guard 4, Probe D & Task E (Inductive Typestate Trace Safety, Absorbing `ZEROED_FREED` & Differential Trace Equivalence):** Theorems 1–5 prove over arbitrary event lists `evs : List LifecycleEvent` that `ZeroedFreed` is strictly absorbing and {`AbortedPoisoned`, `ZeroedFreed`} never reaches `SealedActive` or `ShutdownClean`, while `test_36` verifies state-for-state equivalence between Python `RequestSlot` and `data/lean_transition_table.json` across 200 random traces (`test_18`, `test_24`, `test_30`, `test_36`).

5. **Guard 5, Probe E & Tasks A/B (Mandatory Pinned-Key Ed25519 Signing, Signed Head Record, Subprocess Isolation & Merkle-DAG):** `Ed25519AuthoritySigner` requires `cryptography` and an explicit seed (rejecting unconfigured or retired `v1.2.0` seeds, `test_28`, `test_35`), `verify_chain_list` requires keyword-only `expected_pubkey` and verifies `signed_head` against tail truncation (`test_26`, `test_35`), `IsolatedSignerSubprocess` isolates or generates the signing key in a child OS process (`test_32`), and `merge_divergent_ledgers` satisfies $(A \sqcup B) \sqcup C = A \sqcup (B \sqcup C)$ (Theorem 12, `test_19`, `test_25`).

6. **Track C & Task F (Illustrative 120-Segment Piecewise Duhamel Parameter Study on `data/wp19_segments.json`):** Theorem 15 (`duhamel_piecewise_contraction_bound`) proves $\alpha(\alpha e_0 + d_1) + d_2 < G(Ge_0 + d_1) + d_2$ for $0 \le \alpha < G$ and $d_1 > 0$, and `evaluate_wp19_piecewise_duhamel_study` evaluates the 120-segment recurrence in 200-bit `mpmath.iv` arithmetic using the SHA-256-verified parameters in `data/wp19_segments.json` (10.5281/zenodo.23034545, `test_33`, `test_34`).

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